Status	Finished
Started	Tuesday, 13 October 2026, 9:49 AM
Completed	Tuesday, 13 October 2026, 9:57 AM
Duration	8 mins 28 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Quick sort

b. Heap sort

c. Selection sort

d. Merge sort

The correct answer is:
Merge sort

Question **2**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Linear Search
b. Binary Search
c. Bubble Sort
d. Merge Sort

The correct answer is: Merge Sort

Question **3**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Merge Sort
b. Bubble Sort
c. Quick Sort
d. Heap Sort

The correct answer is: Bubble Sort

Question **4**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Merge Sort
b. Insertion Sort
c. Quick Sort
d. Bubble Sort

The correct answer is: Bubble Sort

Question **5**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Deleting data from a list
b. Arranging data in a particular format
c. Inserting data into a list
d. Searching for data in a list

The correct answer is: Arranging data in a particular format

Question **6**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Combining sorted sublists
b. Dividing the input into smaller subproblems
c. Comparing each element with the others
d. Sorting the entire list sequentially

The correct answer is: Dividing the input into smaller subproblems

Question **7**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Divide and conquer
b. Greedy method
c. Backtracking
d. Dynamic programming

The correct answer is:
Divide and conquer

Question **8**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Sorting the entire list in one pass
b. Bubbling up the smallest element
c. Dividing the list into halves
d. Bubbling up the largest element to its correct position

The correct answer is: Bubbling up the largest element to its correct position

Question **9**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Rearranging
b. Searching
c. Merging
d. Sorting

The correct answer is: Sorting

Question **10**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Distribution

b. Insertion

c. Selection

d. Extraction

The correct answer is:
Insertion

Question **11**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It increases the number of comparisons needed

b. It allows for faster searching

c. It has no effect on the search operation

d. It makes the search operation slower

The correct answer is: It allows for faster searching

Question **12**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It is based on the divide-and-conquer approach
b. It does not require recursion
c. It sorts data using a single pass
d. It is less efficient than bubble sort

The correct answer is: It is based on the divide-and-conquer approach

Question **13**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It is less efficient than custom sorting algorithms
b. It requires external libraries
c. It only works with integer arrays
d. It sorts data out of the box efficiently

The correct answer is: It sorts data out of the box efficiently

Question **14**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Searching

b. Complexity

c. Merging

d. Sorting

The correct answer is:
Complexity

Question **15**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It compares adjacent elements and swaps them if necessary

b. It builds a sorted array one element at a time

c. It repeatedly finds the minimum element and moves it to the sorted part of the list

d. It divides the list into two halves, sorts each half, and then merges them

The correct answer is: It divides the list into two halves, sorts each half, and then merges them